

Assessments of the Potential for Integrating Battery Energy Storage System into Vietnam National Grid

The-Anh Kieu
School of Electrical and
Electronic Engineering
Hanoi University of Science and
Technology
Hanoi, Vietnam
anh.kt221878@sis.hust.edu.vn

Nhung Nguyen-Hong
School of Electrical and
Electronic Engineering
Hanoi University of Science and
Technology
Hanoi, Vietnam
nhung.nguyenhong1@hust.edu.vn

Huy Nguyen-Duc
School of Electrical and
Electronic Engineering
Hanoi University of Science and
Technology
Hanoi, Vietnam
huy.nguyenduc1@hust.edu.vn

Khanh Tran-Van
Northern Electrical Testing
Company limited
Northern Power Corporation -
EVN
Hanoi, Vietnam
khanhtrantrn3@gmail.com

Abstract—The global energy transition is accelerating with the integration of Renewable Energy Sources (RES) into the power grid, and Vietnam is no exception. However, when integrating RES, the stability and reliability of the grid become issues that need to be handled. Vietnam's grid currently lacks sources that can respond quickly enough to the uncertainty of RES. Therefore, integrating a fast-response system such as Battery Energy Storage System (BESS) is worth considering. This research focuses on assessing the impact of installing the BESS on operational costs and reliability through the Loss of Load Probability (LOLP) and Loss of Load Expectation (LOLE) for Vietnamese power system. The Unit Commitment optimization problem model is formulated as a mixed linear integer programming (MILP) model and solved with PyPSA/CPLEX software. The power system's reliability model is also presented. Data were gathered from the operation and parameters of actual generator units in Vietnam's power system. The results indicate that BESS utilization enhances flexibility, optimizes the use of RES, and reduces the need for traditional sources, thus lowering costs and increasing the reliability of the power system. This study is expected to provide valuable information for planning and investment in this field.

Keywords—battery storage system, LOLE, LOLP, renewable energy resources, unit commitment

I. INTRODUCTION

In recent years, countries around the world, including Vietnam, have been focusing on developing RES, such as wind power plants (WP) and photovoltaic farms (PV), to reduce fossil consumption and CO₂ emissions [1]. Report [2] shows the great potential of RES in Vietnam. Solar power has the most significant potential, with a capacity of up to 500 GW. Wind power is the second largest RES, with more than 221 GW of potential onshore wind power capacity [3]. Furthermore, Vietnam's Eighth National Power Development Plan also targets installing 6 GW of offshore wind power by 2030, and this capacity is expected to increase at least 10 times compared with the current power by 2050 [4]. In addition, the report [5] shows that Vietnam needs to install an additional 7,000 - 8,000 MW of capacity each year to meet the power system load's growth rate and the country's modernization goals. This has led to the role of RES in Vietnam becoming more important.

However, as the penetration of RES sources increases, the grid's reliability is also a matter of concern. For example, the uncertainty of RES can lead to a power imbalance, and then the system's frequency can fluctuate outside the allowable operating range. Therefore, it is necessary to integrate BESS into the power system. The significant advantage of BESS is its fast-

response ability to compensate for the power imbalance in real-time. In addition, many studies show that if a BESS is scheduled efficiently, it can reduce the power system's operating cost [6], [7], [8], [9]. Report [10] shows that BESS has a very high growth rate of installed capacity worldwide. Notably, lithium-ion batteries are widely used due to advances in research, development, and economics. Lithium-ion batteries in 2023 are about 140 USD/MWh, much lower than 1,200 USD/MWh in 2010 [11]. This is one of the fastest cost reductions compared to any technology in history.

Vietnam's Eighth National Power Development Plan also mentions the integration of BESS in Vietnam's power grid. However, BESS is a type of investment that has yet to appear in Vietnam, so it is necessary to calculate its installation to consider its accuracy, economics, and possibility. This paper assesses the impact of BESS on the Vietnamese power system's operating costs. The problem is formulated as a mixed linear integer programming (MILP) model, including RES, BESS, and grid constraints, and solved by PyPSA/CPLEX. Data in this paper is collected from the actual operation of the Vietnamese power system. Then, the authors calculated indicators such as the probability of power shortage with the Loss of Load Probability (LOLP) value and the Loss of Load Expectation (LOLE) to verify the reliability of the electricity power system.

II. PROBLEM FORMULATION

A. Overview

The paper modeled Vietnam's electricity system in a 3-buses structure, including sources such as thermal power (coal and gas), hydropower, photovoltaic power, wind power, a BESS, and the 500kV North-South transmission line system (Figure 1). Some other energy sources, such as electricity imported from Laos and China, are excluded. The output of RES is calculated based on the Power – Solar Irradiance and the Power – Wind Speed characteristics curve [12].

The process of the research is outlined in Figure 2. It begins by adjusting the input parameters, including Solar Irradiance, Wind Speed datasets collected from the Copernicus Climate Change Services (C3S) [13]. These datasets are converted to output power, combined with data from generators, National Load and BESS. An optimization model is built with an objective function and some technical constraints for each type of participating component. After the UC optimization model, the authors checked the sensitivity of this program by

calculating the LOLP and LOLE values and compared them with the original ones.

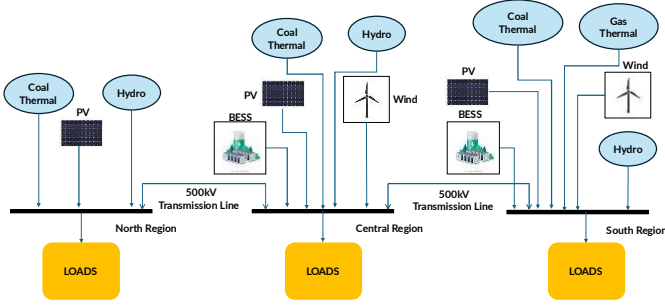


Fig. 1. Structure of the Vietnamese National Electricity Grid model

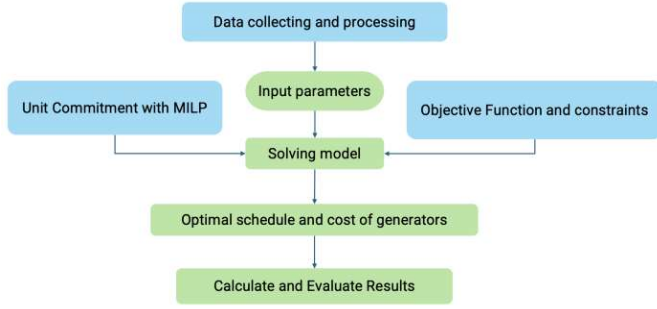


Fig. 2. Research Process Flowchart

B. Optimization Model

In this section, the scheduling operation of power generators—the Unit Commitment (UC) problem—is introduced to calculate the optimal operation for the Vietnam Electricity System to minimize the total cost of all generators. The model maximizes the use of RES and leverages BESS's flexible charging/discharging capability to meet the demand.

Assuming the marginal cost of the wind, solar power systems and BESS have a very low marginal cost. Then, the objective function of the model is formulated as follows:

$$\text{Min} \sum_{t=1}^T \sum_{g=1}^G C_g \cdot P_{g,t} + \sum_{t=1}^T \sum_{i=1}^I (C_{i,t}^{SD} \cdot z_{i,t} + C_{i,t}^{SU} \cdot y_{i,t}) \quad (1)$$

Where T is the total optimization period with the participation of G generator units with different types of power generation. $P_{g,t}$ is the dispatched power of unit g in the time interval t , with the marginal cost C_g .

In addition, the optimization model needs to satisfy the following technical constraints:

- Thermal Generator constraints

Constraints (2), (3), and (4) ensure that each plant's output stays within permissible limits. Additionally, startup and shutdown actions are limited by technical and cost constraints, represented in constraints (5) and (6) by minimum uptime TU_i and minimum downtime TD_i .

$$u_{i,t} \cdot P_i^{min} \leq P_{i,t} \leq u_{i,t} \cdot P_i^{max} \quad (2)$$

$$P_{i,t} - P_{i,t-1} \leq RU_i \cdot u_{i,t-1} \quad (3)$$

$$P_{i,t-1} - P_{i,t} \leq RD_i \cdot u_{i,t} \quad (4)$$

$$\sum_{t'=t}^{t+TU_i} u_{i,t'} \geq TU_i \cdot (u_{i,t} - u_{i,t-1}) \quad (5)$$

$$\sum_{t'=t}^{t+TD_i} (1 - u_{i,t'}) \geq TD_i \cdot (u_{i,t-1} - u_{i,t}) \quad (6)$$

- Hydro Generator constraints

The energy output from hydropower plants must not exceed the scheduled dispatch for the day or week cycle, denoted by T_A .

$$\sum_t^{T_A} u_{h,i} \cdot P_{h,i} \leq A_h^{schedule} \quad (7)$$

$$u_{h,t} \cdot P_h^{min} \leq P_{h,t} \leq u_{h,t} \cdot P_h^{max} \quad (8)$$

- RES operational constraints

The available power from RES is calculated from characteristic curves and must remain within limits.

$$0 \leq u_{w,t} \cdot P_{w,t} \leq u_{w,t} \cdot P_{w,t}^{max} \quad (9)$$

$$0 \leq u_{s,t} \cdot P_{s,t} \leq u_{s,t} \cdot P_{s,t}^{max} \quad (10)$$

- BESS constraints

Constraints (11) and (12) specify charging and discharging limits within inverter capabilities. Battery State-of-Power after each time interval depends on the previous value, charge/discharge power, and an energy loss parameter η_b^{loss} . The energy stored in BESS is limited by its rated capacity. Besides, the BESS energy should return to the initial set power after each operating duration.

$$0 \leq P_{b,t}^{ch} \leq u_{b,t} \cdot P_b^{max} \quad (11)$$

$$0 \leq P_{b,t}^{dch} \leq (1 - u_{b,t}) \cdot P_b^{max} \quad (12)$$

$$E_{b,t} = (1 - \eta_b^{loss}) \cdot E_{b,t-1} + \eta_b^{ch} \cdot P_{b,t}^{ch} - \frac{P_{b,t}^{dch}}{\eta_b^{dch}} \quad (13)$$

$$0 \leq E_{b,t} \leq E_b^{rated} \quad (14)$$

$$E_{b,t=0} \leq E_{b,t=24} \quad (15)$$

- Power balance constraints

The total power output of generations and BESS must balance the system load at any time t .

$$\sum_g P_{g,t} + \sum_b (P_{b,t}^{dch} - P_{b,t}^{ch}) = P_{load,t} \quad (16)$$

- Transmission Line constraints

Power flow across transmission lines must satisfy the maximum transmission limits in both directions.

$$-P_l^{max} \leq P_{l,t} \leq P_l^{max} \quad (17)$$

C. LOLP and LOLE Calculation

LOLP and LOLE are common metrics used to assess the reliability of power systems. There are multiple methods to calculate these indicators, and the approach in this study is described as follows:

For the LOLP indicator, the unavailability factor Forced Outage Rate (FOR) is calculated first using the formula (18):

$$FOR = \frac{\text{Forced Outage Time}}{\text{Operating Time} + \text{Forced Outage Time}} \quad (18)$$

From (18) we can calculate the system's availability in (19):

$$p = 1 - FOR \quad (19)$$

Next, a Capacity Outage Probability Table (COPT) is constructed, where combinations of power units in available and unavailable states are ordered. The COPT takes into account the availability of generation units to determine the overall system availability and unavailability. The states in the table are organized in ascending order based on the combined capacity of the generating units.

For a system with n generators, the LOLP index is calculated by (20):

$$LOLP = \sum_{i=1}^n p_i t_i \quad (20)$$

In (20), p_i is the power outage probability of the generator, t_i is the power outage period according to the load graph in percentage.

Following (21), the LOLE indicator is calculated by multiplying the expected outage duration by the cumulative probability of failure for each relevant state.

$$LOLE = \sum_{i=1}^n p_i T_i \quad (21)$$

where p_i is the probability of power state i , T_i is the outage duration in days or hours.

After setting up the optimization model, the LOLP and LOLE values were determined using a Python-based model. This model employs the K-means clustering algorithm [14] to

group similar units into 10 clusters for each bus in the system. Calculations are performed hourly and repeated over 8,760 hours to obtain annual results.

III. RESULTS AND DISCUSSION

A. Input Parameters

The model uses a 3-bus structure for Vietnam's electricity system, representing the North, Central, and South regions. This system includes 672 generator units with a total installed capacity of 72,825 MW. The authors modeled two 500 kV transmission lines in Vietnam with 2,000 MW and 4,000 MW transmission limits. Besides, the authors propose integrating BESS in the Central and Southern territories, where RES is invested and developed significantly due to favorable conditions.

Thermal power units have rated capacity ranging from 4.2 MW to 1,000 MW, with marginal cost between 900 and 1,150 USD/MWh. For RES, wind speed and solar irradiance data were taken from the ERA5 dataset, as introduced in section II.A.

Load data collected from 2020 with an hourly resolution was scaled up to the 2022 version based on growth rates outlined in [15] to serve the optimal operation problem.

The proposed BESS investment capacity is 2,000MWh. In addition, other parameters of the elements in the system are shown in Table 1.

TABLE I. INPUT PARAMETERS

Parameters	Value
Minimum generation time (hours)	8
Minimum shut down time (hours)	3
Ramp up/down rate limits (%)	30
Start-up cost for thermal units (USD)	200
Shut-down cost for thermal units (USD)	150
BESS charging/discharging efficiency (%)	98
BESS Loss Factor (%)	0.5
BESS Inverter Power Rating (MW)	500
Maximum charging time (hours)	4

B. Case Studies and Results

After processing the input data, the authors ran the model on PyPSA / CPLEX software with an optimal calculation time of 8,760 hours, and a calculation cycle of 1 week. The results were calculated according to 2 scenarios:

- Scenario 1: The system does not integrate BESS
- Scenario 2: The system invests in BESS to participate in the power system

To facilitate the analysis of the results, the power system operation plan for a typical week (week 26) corresponding to the two above scenarios is presented in Figures 3 and 4.

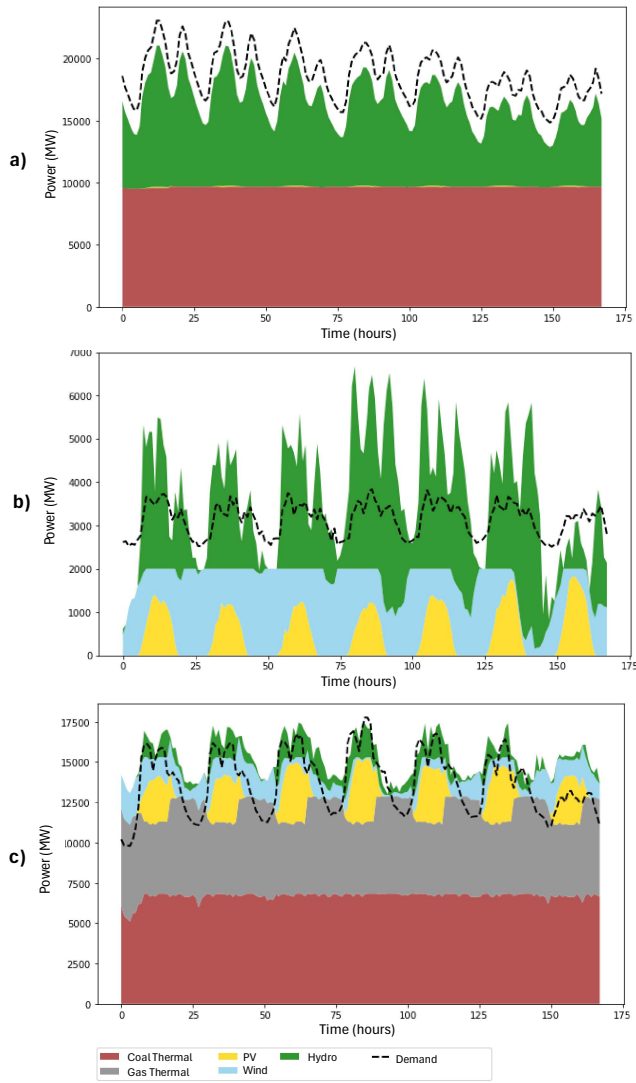


Fig. 3. The Vietnam National Grid's Optimization Schedule without BESS
a) Northern Region; b) Central Region; c) Southern Region

The results in Figures 3 and 4 show that the penetration level of RES in scenario 2 is higher than in scenario 1. In detail, 23,749 MWh of coal-fired thermal power in week 26 can be replaced by RES based on the operation of BESS (Table 2). Then, operating costs for both scenarios were calculated, showing that with BESS, the power system would save 979,690 USD in that week.

TABLE II. POWER COMPARISON OF EACH TYPES OF GENERATORS

	Scenario 1 (MWh)	Scenario 2 (MWh)	Deviation Δ (MWh)
Coal Thermal	2,748,172	2,724,423	23,749
Gas Thermal	896,013	895,857	156
Hydro Power	1,639,236	1,639,753	-517
Wind Power	395,159	398,044	-2,884
PV Power	300,825	322,985	-22,160

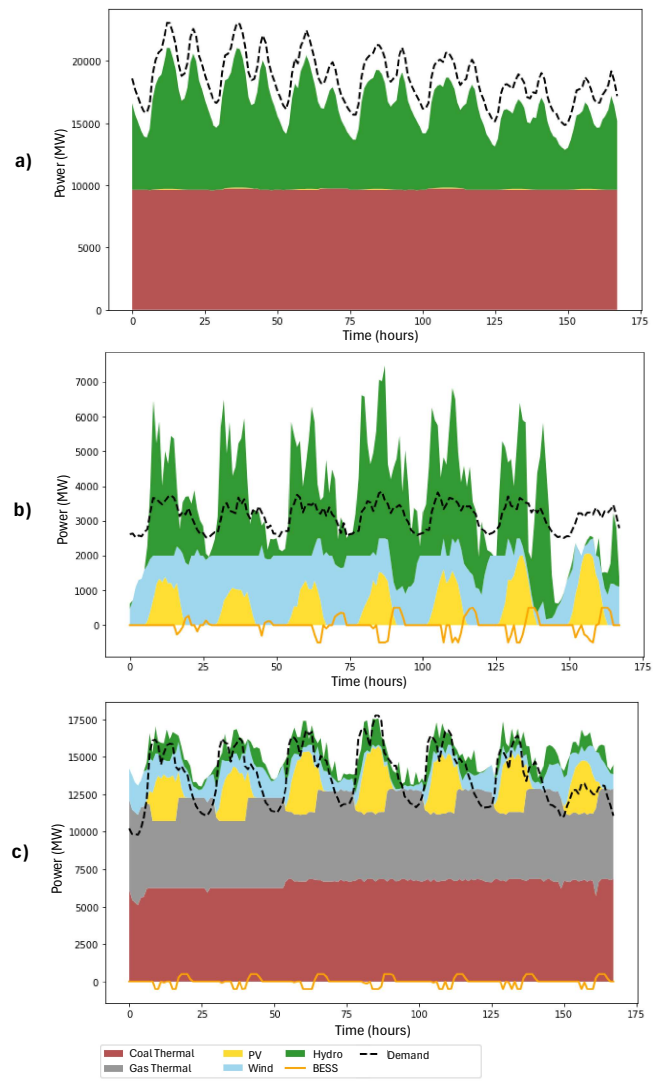


Fig. 4. The Vietnam National Grid's Optimization Schedule with BESS
a) Northern Region; b) Central Region; c) Southern Region

In addition, Figure 5 shows BESS's charging/discharging status in a week, indicating a pretty good penetration into grid operations. BESS usually stores energy between 12:00 and 17:00 each day, when PV units reach their maximum, preventing overload and intra-regional congestion.

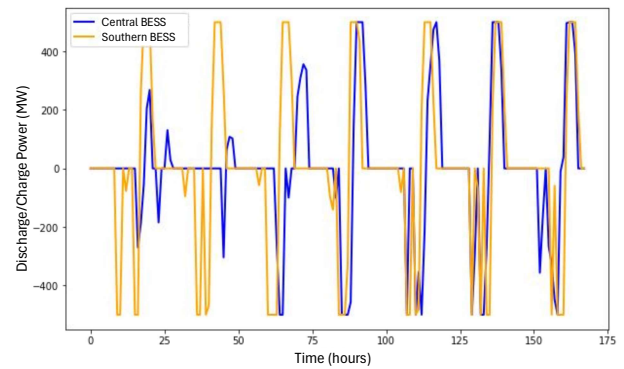


Fig. 5. BESS's weekly operation

TABLE III. OPERATION COST IN A YEAR OF VIETNAM NATIONAL GRID

	Operation Cost (Million USD)
Scenario 1	7766
Scenario 2	7715
Deviation Δ	51

The authors ran the optimization model over 52 consecutive weeks to optimize annual operations; then synthesized the annual operating costs of the two scenarios and obtained the results in Table 3. It can be seen that due to the presence of BESS, the operating cost of scenario 2 is less than that of scenario 1, the difference in cost between the two scenarios is 51 million USD. Besides, the authors tested many cases with load growth rates from 1.05 to 1.20 and with different BESS capacities, from 2,000 MWh to 3,000 MWh. The results indicate that the benefits are still guaranteed, with the difference in operating costs between the two scenarios in 1 week ranging from approximately 900,000 USD to 1,200,000 USD. Assuming the installation of two 500 MW / 2,000 MWh BESS units with a lifespan of 15-20 years (according to NREL [16]) and an investment starting in 2025 for 480 million USD (120,000 USD/MWh), the payback period would be approximately 10 years. This analysis highlights the effectiveness and viability of BESS investments both now and in the future.

C. LOLP and LOLE Calculation

After running the optimization model, the authors examined the reliability indicators LOLP and LOLE. The results are displayed in Table 4. The results demonstrate that investing in BESS helps increase the system's reliability by reducing the number of power outages per year, thereby seeing its effectiveness in societal benefits.

TABLE IV. LOLP AND LOLE VALUES IN 2 SCENARIOS

	LOLP (%)	LOLE (hours)
Without BESS	0.4680	41
With BESS	0.3425	30

IV. DISCUSSION

This paper studies the impact of BESS integration into the Vietnamese electricity power system. The authors created two scenarios, one when BESS was not integrated and the other when BESS was installed in the central and southern areas, to compare operating costs and LOLP/LOLE indexes. The impact of load growth rate and BESS sizing on the results is also analyzed.

The results show that if BESS is integrated into the power system, we can reduce the power output of thermal power plants and fully exploit the RES's available power output. This will consequently actively benefit the environment and gradually lead to zero net CO₂ emissions by 2050. Along with the different schemes of load growth rate or storage capacity mentioned above, economic profit and reliability are always guaranteed.

The model still has a lot of potential for future development. In addition to calculating optimal operation scheduling, the model can still be used to check the stability speed of the power system frequency when there are parameters of renewable energy sources, thereby further increasing the reliability of the power system.

REFERENCES

- [1] IEA, "Tracking Clean Energy Progress 2023 – Analysis," IEA, Jul. 2023. <https://www.iea.org/reports/tracking-clean-energy-progress-2023>
- [2] Department of Electricity and Renewable Energy - Ministry of Industry and Trade, Danish Energy Agency, Institute of Energy et al., "Vietnam Energy Outlook Report," 2024.
- [3] N. Online, "Prospects and Barriers for Wind Power Development Globally and in Vietnam in the Short and Medium Term," Vietnam Energy Online, Jul. 09, 2024. <https://nangluongvietnam.vn/trien-vong-rao-can-phat-trien-dien-gio-tren-the-gioi-viet-nam-trong-ngan-va-trung-han-32826.html>
- [4] Vy, "Outlook positive for Vietnam's wind power," *Vietnam Investment Review - VIR*, Apr. 23, 2024. <https://vir.com.vn/outlook-positive-for-vietnams-wind-power-110588.html>
- [5] B. C. Thuong, "National Power Development Plan for the Period 2021 - 2030, with a Vision to 2050," 2023, [Online]. Available: <https://xaydungchinhsach.chinhphu.vn/toan-van-quy-hoach-phat-trien-dien-luc-quoc-gia-11923051616315244.htm>
- [6] N. Chatrungs, "Battery Energy Storage System (BESS) and Development of Grid Scale BESS in EGAT," 2019 IEEE PES GTD Gd. Int. Conf. Expo. Asia, GTD Asia 2019, pp. 589–593, May 2019, doi: 10.1109/GTDASIA.2019.8715953.
- [7] M. T. Lawder et al., "Battery energy storage system (BESS) and battery management system (BMS) for grid-scale applications," *Proc. IEEE*, vol. 102, no. 6, pp. 1014–1030, 2014, doi: 10.1109/JPROC.2014.2317451.
- [8] S. Bin Wali et al., "Battery storage systems integrated renewable energy sources: A biblio metric analysis towards future directions," *J. Energy Storage*, vol. 35, p. 102296, Mar. 2021, doi: 10.1016/J.EST.2021.102296.
- [9] Issoufou Tahirou Halidou, Harun Or Rashid Howlader, Mohammed E. Lotfy, Atsushi Yona, and Tomonobu Senjyu, "Unit Commitment in the Presence of Renewable Energy Sources and Energy Storage System: Case Study," *J. Energy Power Eng.*, vol. 12, no. 6, Jun. 2018, doi: 10.17265/1934-8975/2018.06.004.
- [10] IEA (2023), Global installed grid-scale battery storage capacity in the Net Zero Scenario, 2015-2030, IEA, Paris <https://www.iea.org/data-and-statistics/charts/global-installed-grid-scale-battery-storage-capacity-in-the-net-zero-scenario-2015-2030>, Licence: CC BY 4.0.
- [11] "Lithium battery pack prices go up in BloombergNEF annual survey." <https://www.energy-storage.news/lithium-battery-pack-prices-go-up-for-first-time-since-bloombergnef-began-annual-survey/> (accessed Jul. 29, 2023).
- [12] H. Nguyen-Duc, K. Tran-Van, K. Nguyen-Minh and T. -A. Kieu, "Optimal Storage Planning for an Islanded Power System," 2023 Asia Meeting on Environment and Electrical Engineering (EEE-AM), Hanoi, Vietnam, 2023, pp. 1-6, doi: 10.1109/EEE-AM58328.2023.10394691.
- [13] "About us | Copernicus," [climate.copernicus.eu](https://climate.copernicus.eu/about-us). <https://climate.copernicus.eu/about-us>
- [14] A. M. Iktun, A. E. Ezugwu, L. Abualigah, B. Abuhaija, and J. Heming, "K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data," *Inf. Sci. (Ny)*, vol. 622, pp. 178–210, Apr. 2023, doi: 10.1016/j.ins.2022.11.139.
- [15] IEA (2024), World Energy Outlook 2024, IEA, Paris <https://www.iea.org/reports/world-energy-outlook-2024>, Licence: CC BY 4.0 (report); CC BY NC SA 4.0 (Annex A)
- [16] W. Cole and A. Karmakar, "Cost Projections for Utility-Scale Battery Storage: 2023 Update," 2023. Available: <https://www.nrel.gov/docs/fy23osti/85332.pdf>